October 14th, 2026

CMS Tracker DPG Workshop

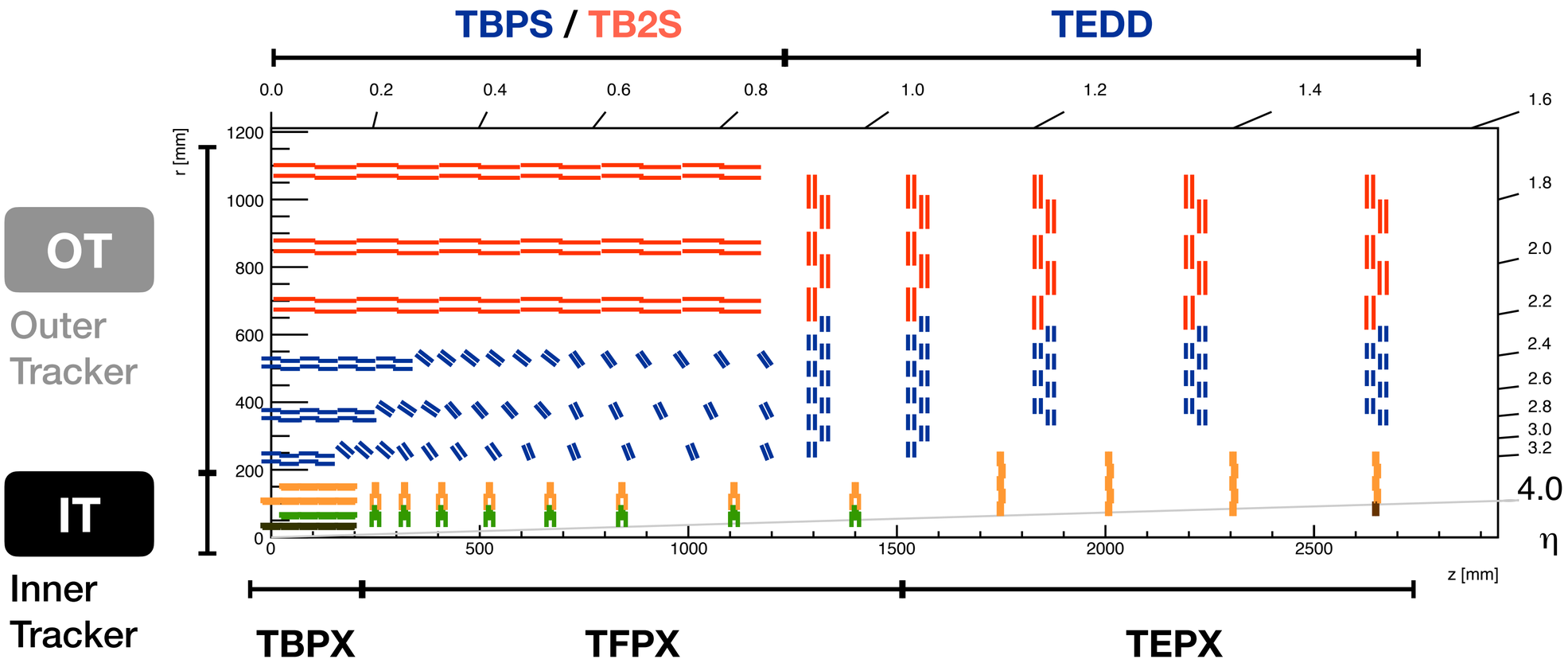
Geometry & Material Budget Overview

N. Bartosik [a, b]

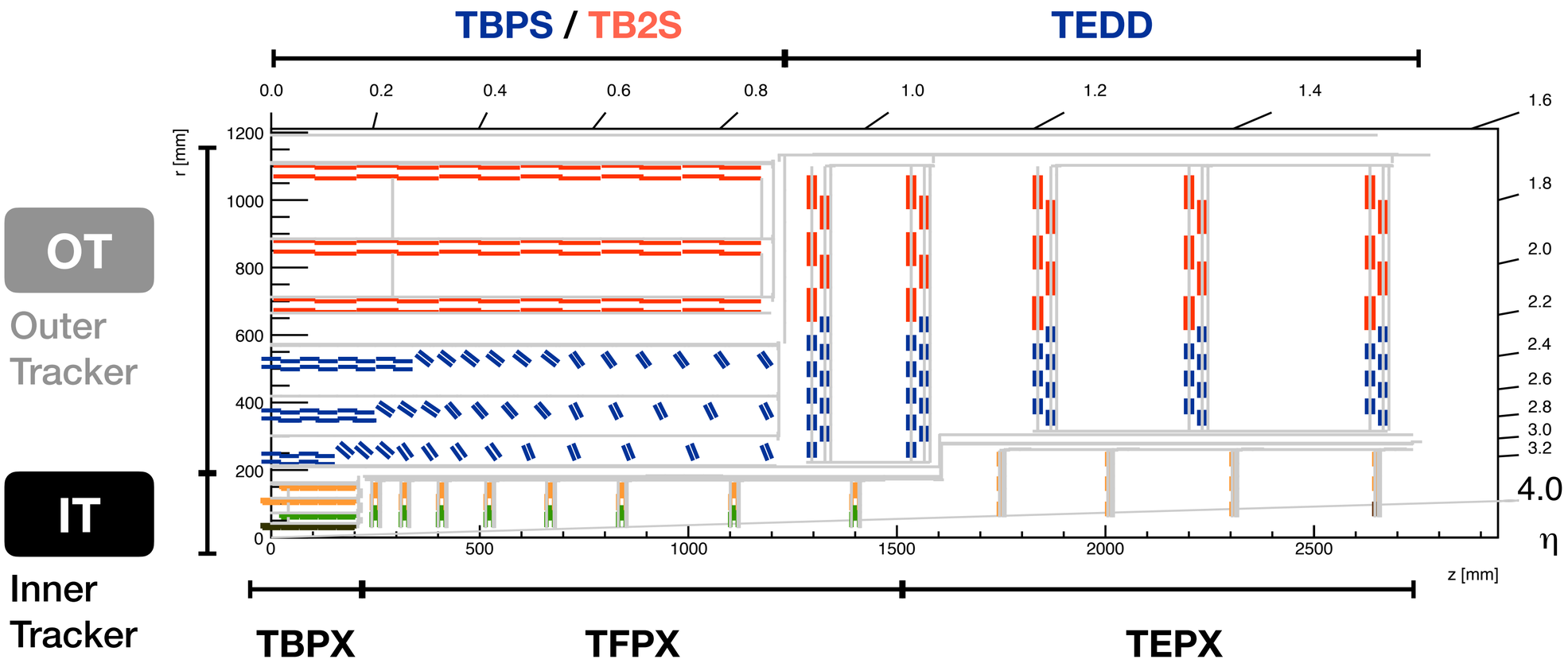
[a] UPO *(Italy)* [b] INFN Torino *(Italy)*

Baseline geometry

Current baseline Tracker geometry: sensitive volumes

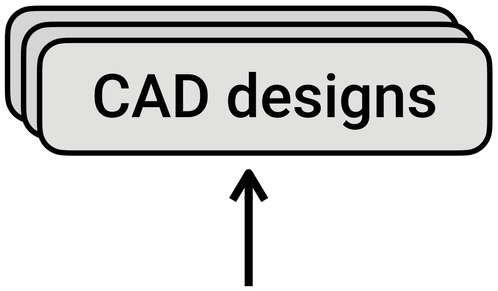


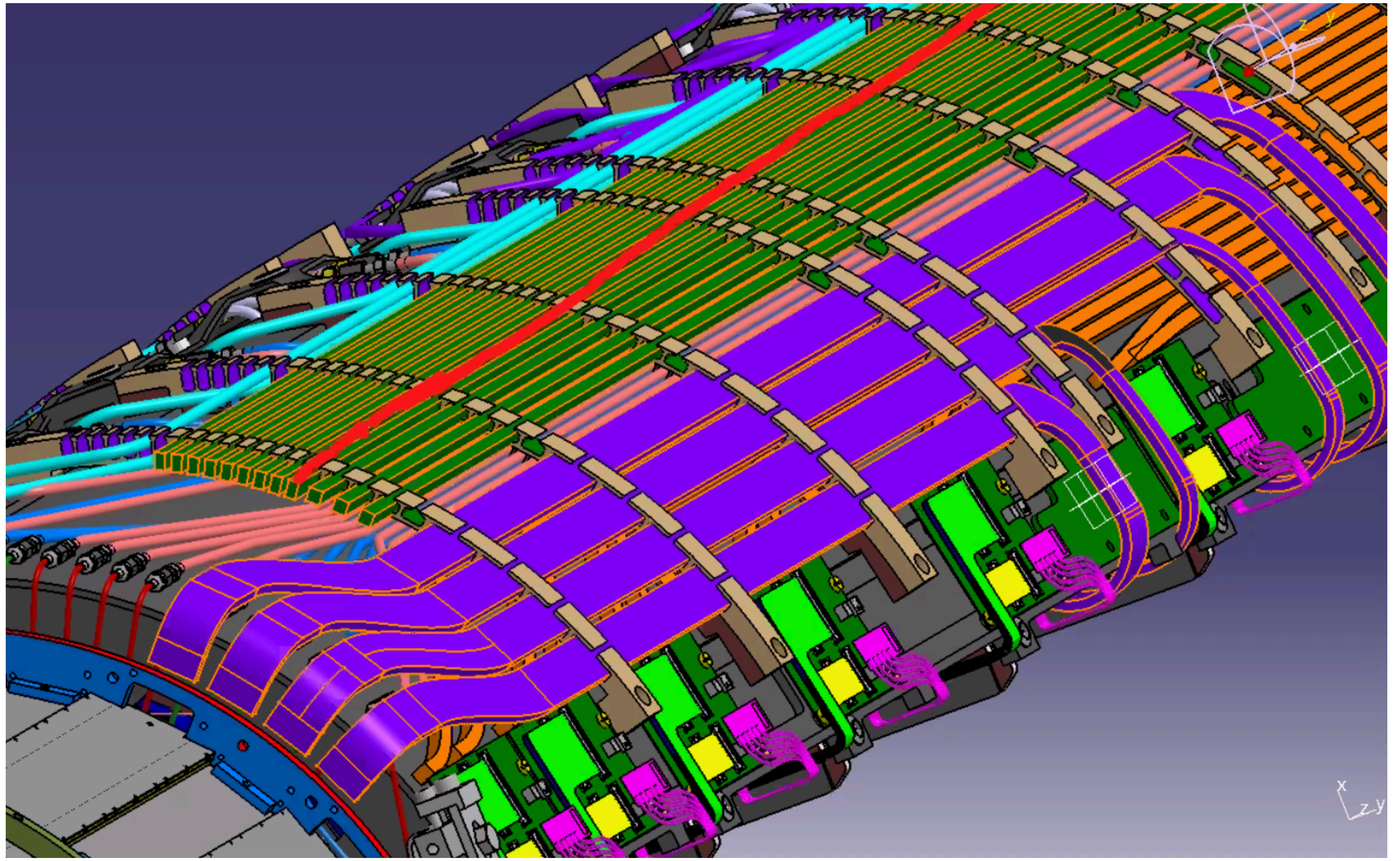
Current baseline Tracker geometry: sensitive volumes **+ passive volumes (services)**



Geometry generation

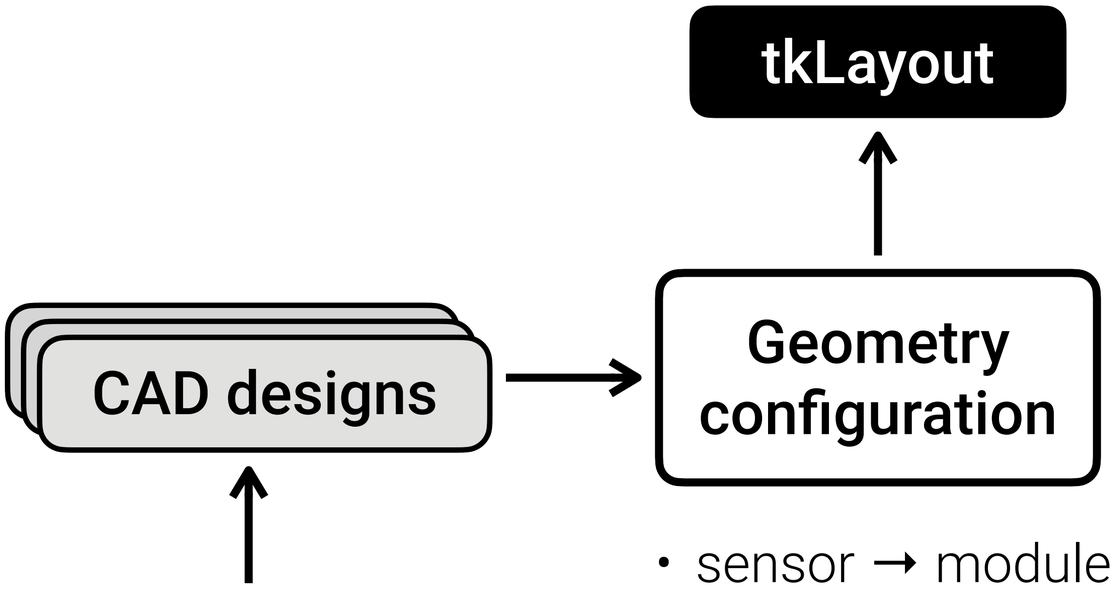
During construction phase CAD drawings are the main source of truth





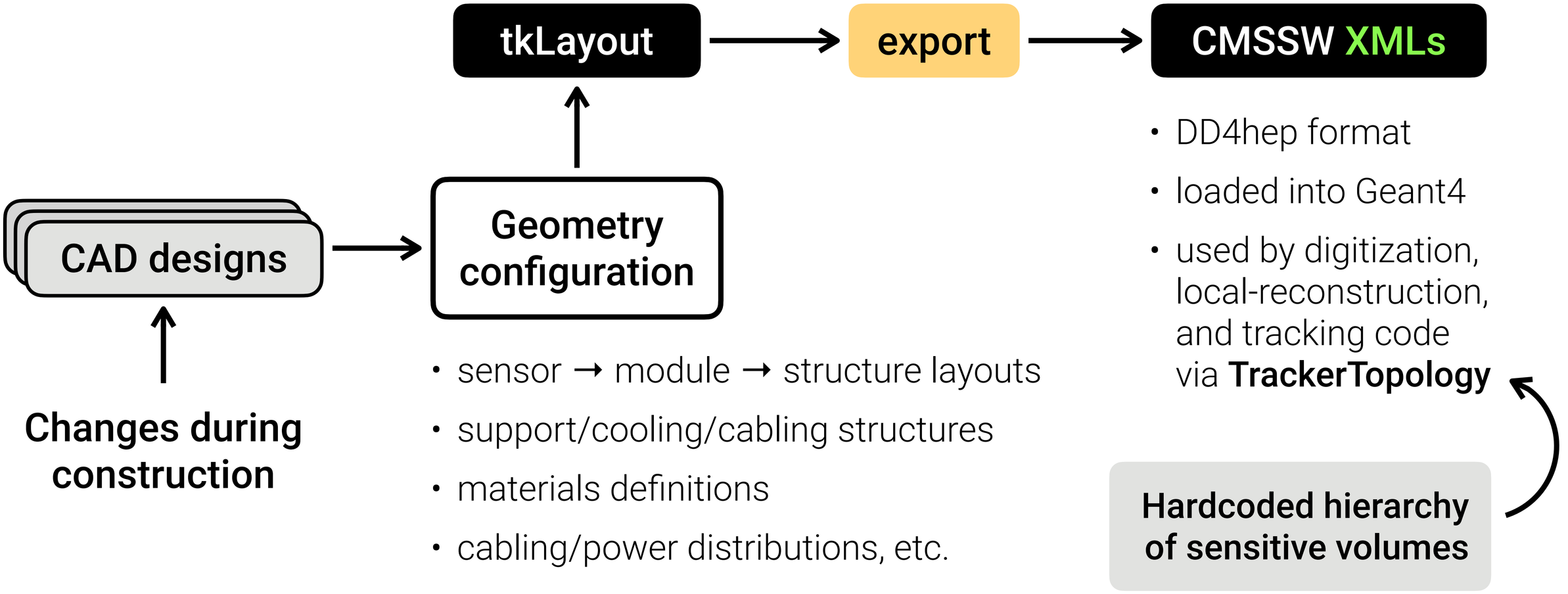
Geometry generation

Geometry defined programmaticaly in tkLayout — external tool with high flexibility



- sensor → module → structure layouts
- support/cooling/cabling structures
- materials definitions
- cabling/power distributions, etc.

Geometry exported to CMSSW using a dedicated module in tkLayout



Geometry naming

More fine-grained naming convention in tkLayout than in CMSSW



OT800_IT711 [link]

- **OT 8.0.0**
  - TBPS 8.0.0
  - TB2S 6.1.6
  - TEDD 8.0.0
- **IT 7.1.1**
  - TBPX 7.1.1
  - TFPX 6.1.5
  - TEPX 6.1.3

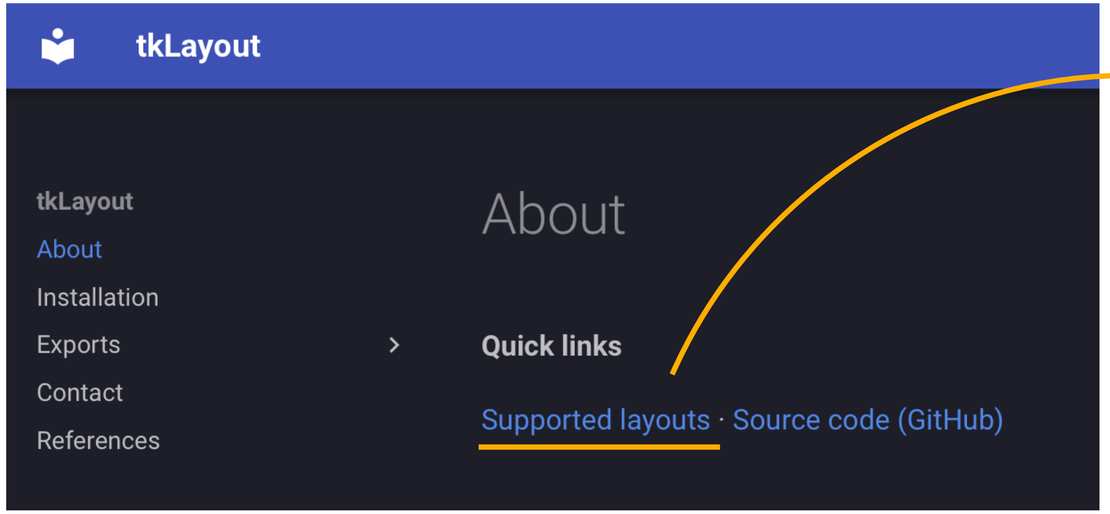
T35 [link]

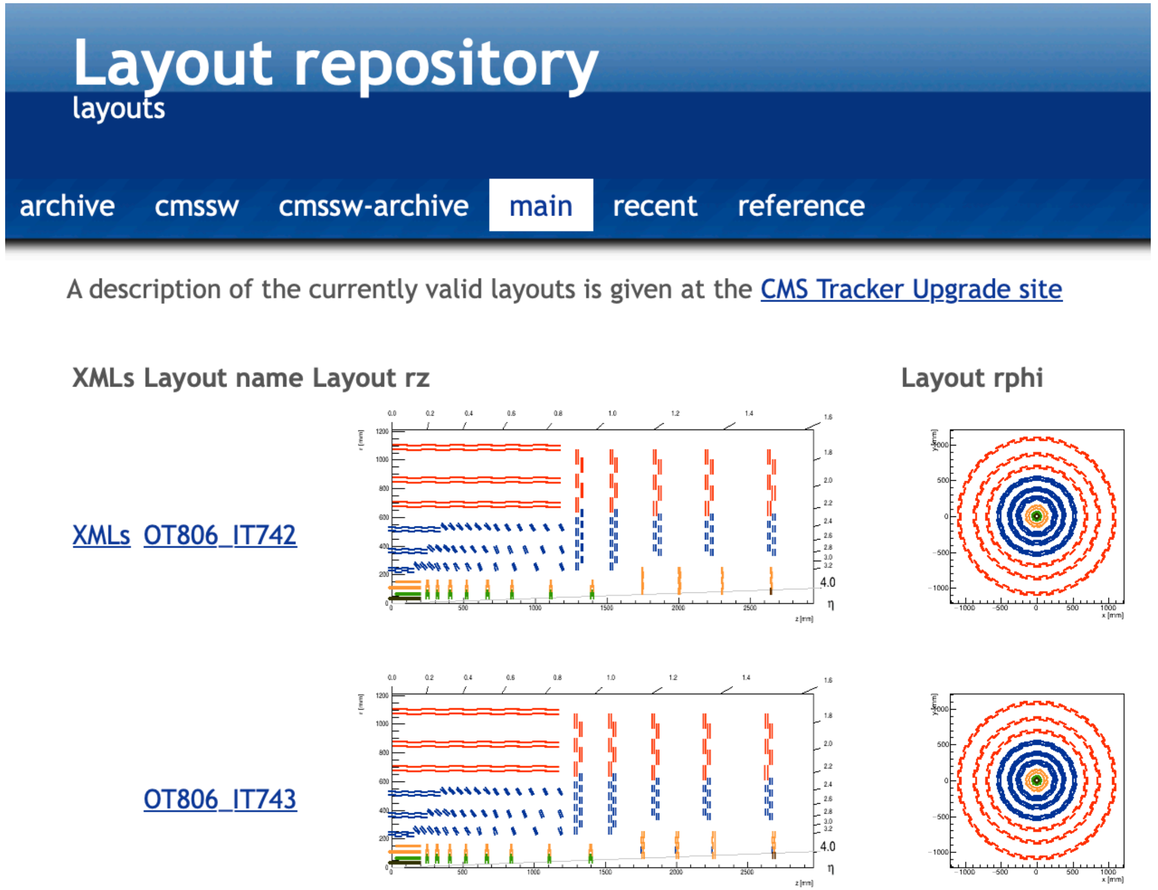
↳ **D128**

T35 + C27 + M16 + I21 + O11 + F10

Geometry of the complete CMS detector

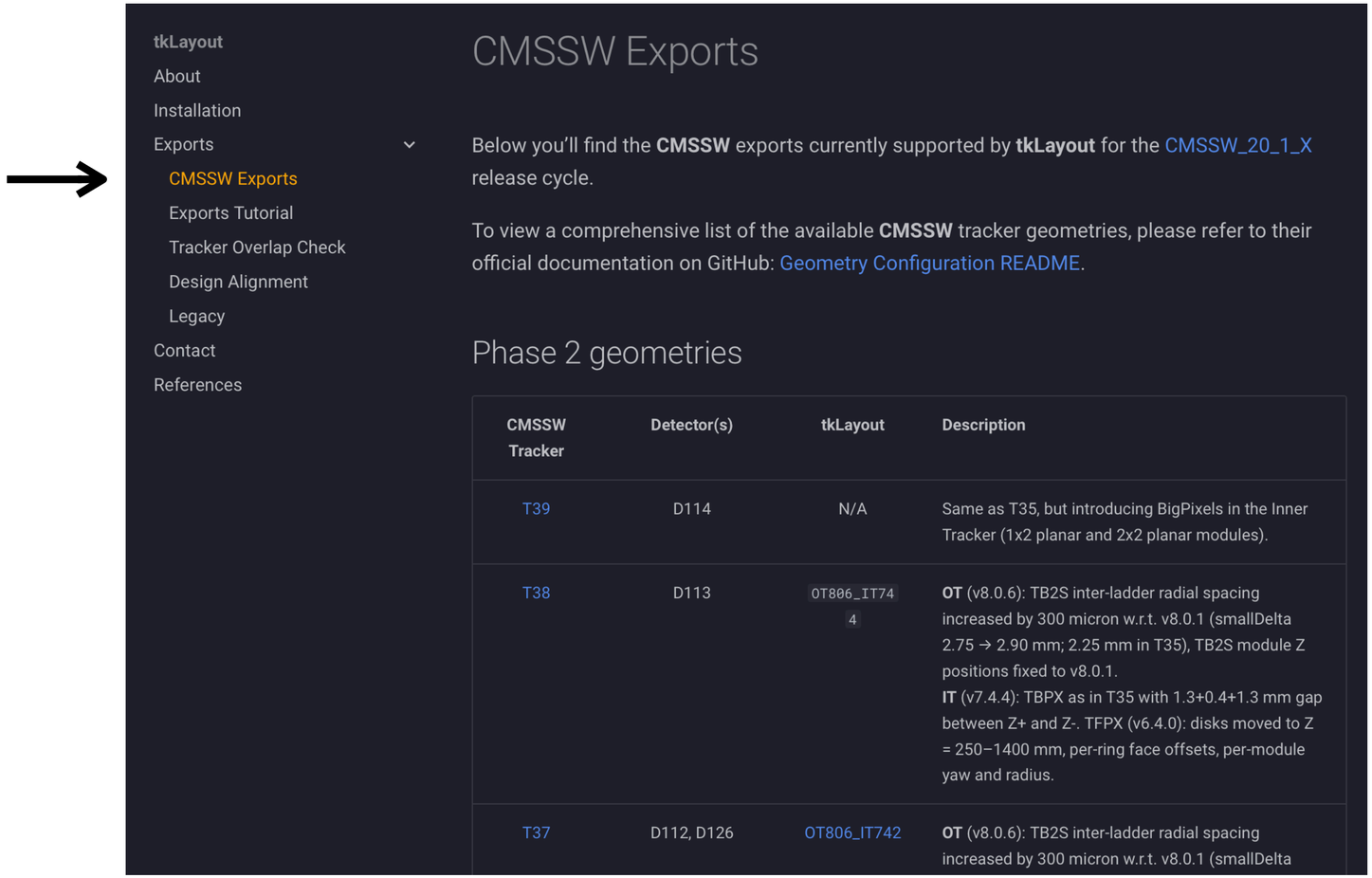
All the relevant information about geometries available from the tkLayout website





Clean-up and minor restructuring of the Layouts page is ongoing

Information about exported geometries listed on the CMSSW Exports page



tkLayout
About
Installation
Exports
CMSSW Exports
Exports Tutorial
Tracker Overlap Check
Design Alignment
Legacy
Contact
References

CMSSW Exports

Below you'll find the **CMSSW** exports currently supported by **tkLayout** for the CMSSW_20_1_X release cycle.

To view a comprehensive list of the available **CMSSW** tracker geometries, please refer to their official documentation on GitHub: Geometry Configuration README.

Phase 2 geometries

CMSSW Tracker	Detector(s)	tkLayout	Description
T39	D114	N/A	Same as T35, but introducing BigPixels in the Inner Tracker (1x2 planar and 2x2 planar modules).
T38	D113	OT806_IT744	**OT** (v8.0.6): TB2S inter-ladder radial spacing increased by 300 micron w.r.t. v8.0.1 (smallDelta 2.75 → 2.90 mm; 2.25 mm in T35), TB2S module Z positions fixed to v8.0.1. **IT** (v7.4.4): TBPX as in T35 with 1.3+0.4+1.3 mm gap between Z+ and Z-. TFPX (v6.4.0): disks moved to Z = 250–1400 mm, per-ring face offsets, per-module yaw and radius.
T37	D112, D126	OT806_IT742	**OT** (v8.0.6): TB2S inter-ladder radial spacing increased by 300 micron w.r.t. v8.0.1 (smallDelta

Geometry documentation

Information about exported geometries listed on the CMSSW Exports page

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T37	D112, D126	OT806_IT742	**OT** (v8.0.6): TB2S inter-ladder radial spacing increased by 300 micron w.r.t. v8.0.1 (smallDelta

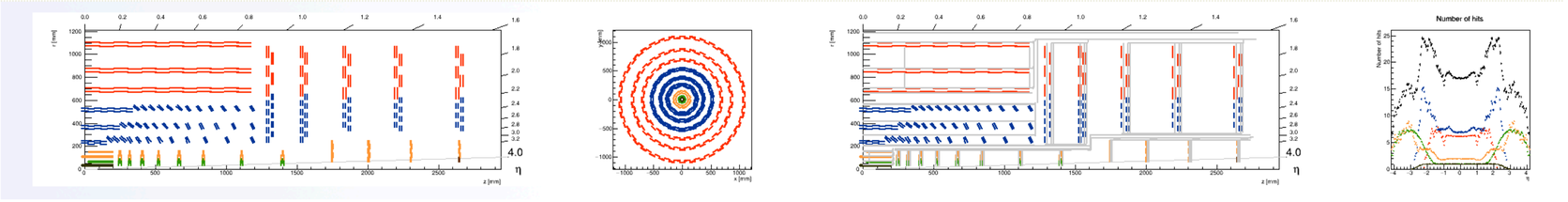
Output of tkLayout simulation

OT806_IT742

layouts

info | geometry | geometry (pixel) | bandwidth | trigger cpus | irradiation (outer) | irradiation (pixelssubdisk) | material (outer) | material (pixel)
weights (pixel) | resolution (pixel) | resolution (tracker) | resolution (trigger) | patternreco | trigger | log page

full layout geometry



Information about exported geometries listed on the CMSSW Exports page

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e available **CMSSW** tracker geometries, please refer to their Geometry Configuration README.

tkLayout	Description
N/A	Same as T35, but introducing BigPixels in the Inner Tracker (1x2 planar and 2x2 planar modules).
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T37	D112, D126	OT806_IT742	**OT** (v8.0.6): TB2S inter-ladder radial spacing increased by 300 micron w.r.t. v8.0.1 (smallDelta

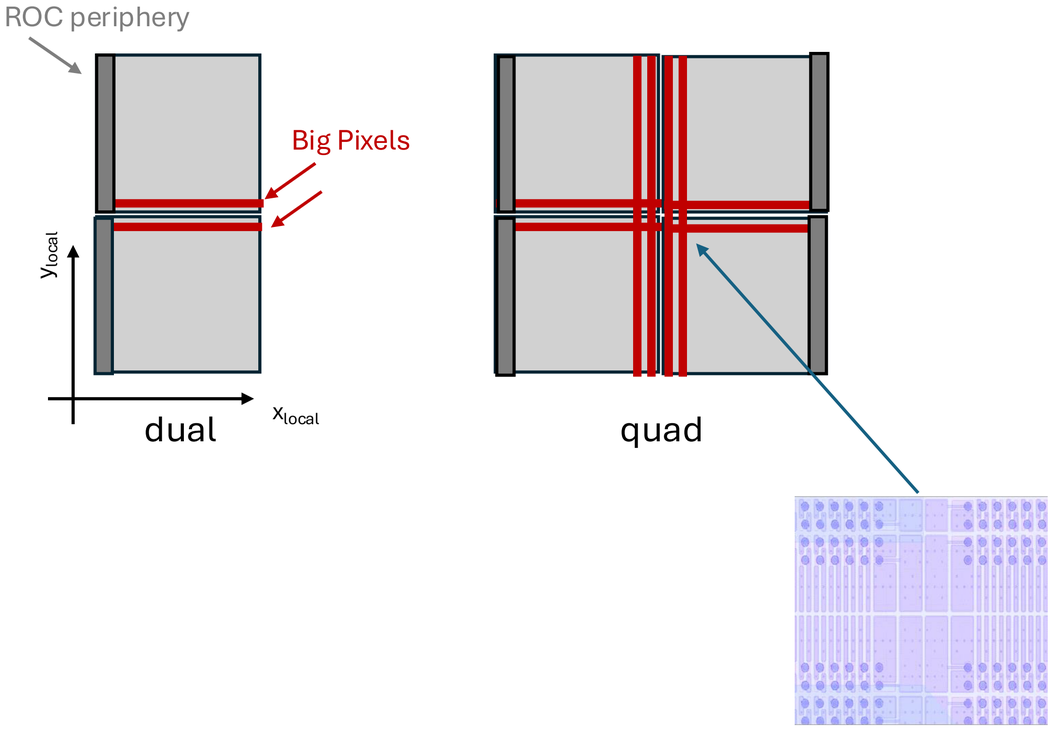
CMSSW definition

```
"T37" : {
    1 : [
        'Geometry/TrackerCommonData/data/PhaseII/TFPXTEPXReordered/trackerParameters.xml',
        'Geometry/TrackerCommonData/data/pixfwdCommon.xml',
        'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_2021_02/pixfwd.xml',
        'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_OT800_IT615_2022_10/pixbar.xml',
        'Geometry/TrackerCommonData/data/trackermaterial.xml',
        'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_OT806_IT741_2024_01/tracker.xml',
        'Geometry/TrackerCommonData/data/PhaseII/OuterTracker616_2020_04/otst.xml',
        'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_OT806_IT742_2024_01/v1/pixel.xml',
        'Geometry/TrackerCommonData/data/PhaseII/TiltedTracker404/trackerbar.xml',
        'Geometry/TrackerCommonData/data/PhaseII/TiltedTracker404/trackerfwd.xml',
        'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_2021_02/trackerStructureTopology.xml',
        'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_OT806_IT741_2024_01/pixelStructureTopology.xml',
        'Geometry/TrackerSimData/data/PhaseII/Tracker_DD4hep_compatible_2021_02/trackersens.xml',
        'Geometry/TrackerSimData/data/PhaseII/Tracker_DD4hep_compatible_OT806_IT741_2024_01/pixelsens.xml',
        'Geometry/TrackerRecoData/data/PhaseII/Tracker_DD4hep_compatible_OT806_IT741_2024_01/trackerRecoMaterial.xml',
        'Geometry/TrackerSimData/data/PhaseII/Tracker_DD4hep_compatible_2021_02/trackerProdCuts.xml',
        'Geometry/TrackerSimData/data/PhaseII/Tracker_DD4hep_compatible_OT806_IT741_2024_01/pixelProdCuts.xml',
        'Geometry/TrackerSimData/data/trackerProdCutsBEAM.xml',
    ],
    "sim" : [
        'from Geometry.TrackerNumberingBuilder.trackerNumberingGeometry_cff import *',
        'from SLHCUpgradeSimulations.Geometry.fakePhase2OuterTrackerConditions_cff import *',
    ],
    "reco" : [
        'from Geometry.CommonTopologies.globalTrackingGeometry_cfi import *',
        'from RecoTracker.GeometryESProducer.TrackerRecoGeometryESProducer_cfi import *',
        'from Geometry.TrackerGeometryBuilder.TrackerAdditionalParametersPerDet_cfi import *',
        'from Geometry.TrackerGeometryBuilder.trackerParameters_cff import *',
        'from Geometry.TrackerNumberingBuilder.trackerTopology_cfi import *',
        'from Geometry.TrackerGeometryBuilder.idealForDigiTrackerGeometry_cff import *',
        'trackerGeometry.applyAlignment = True',
    ],
```

Final adjustments

Next baseline geometry in preparation: **T40**

- **big pixels in IT + OT** (PS-p sensors) **automated in tkLayout**
  IT: fully validated; **OT:** validation ongoing



pixelStructureTopology.xml

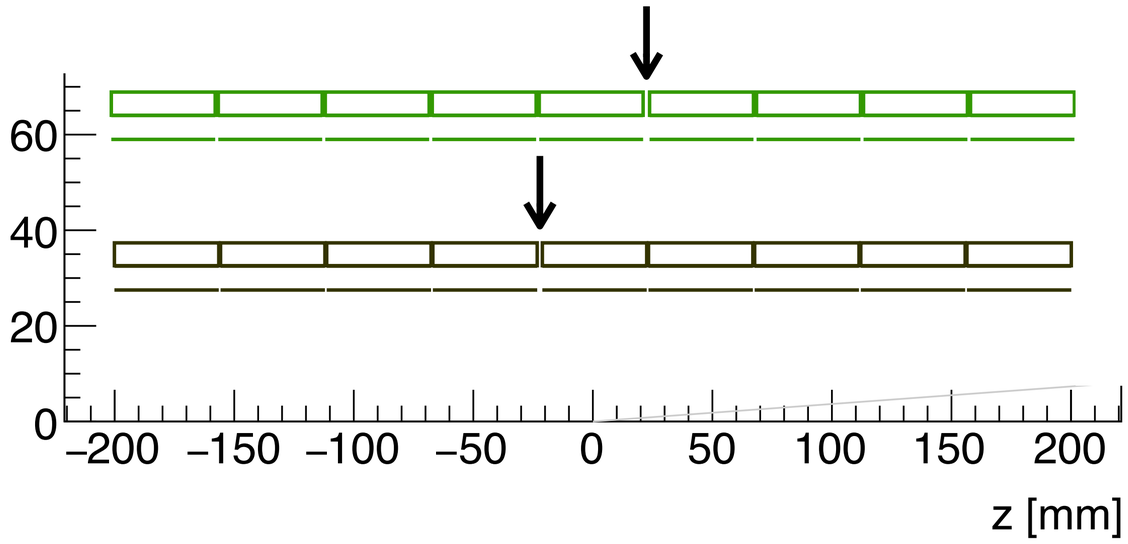
```
<SpecPar name="ITDisc8R2EModuleInnerPixelActiveReadoutPar">
<PartSelector path="//ITDisc8R2EModuleInnerPixelActive"/>
<Parameter name="TkDDDStructure" value="PixelEndcapDet"/>
<Parameter name="PixelROCRows" eval="true" value="672"/>
<Parameter name="PixelROCCols" eval="true" value="216"/>
<Parameter name="PixelROC_X" eval="true" value="1"/>
<Parameter name="PixelROC_Y" eval="true" value="2"/>
<Parameter name="BigPixels_Y" eval="true" value="1"/>
<Parameter name="BigPixels_Pitch_Y" eval="true" value="0.0225"/>
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<PartSelector path="//ITDisc8R3EModuleInnerPixelActive"/>
<Parameter name="TkDDDStructure" value="PixelEndcapDet"/>
<Parameter name="PixelROCRows" eval="true" value="672"/>
<Parameter name="PixelROCCols" eval="true" value="216"/>
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<Parameter name="PixelROC_Y" eval="true" value="2"/>
<Parameter name="BigPixels_X" eval="true" value="2"/>
<Parameter name="BigPixels_Y" eval="true" value="1"/>
<Parameter name="BigPixels_Pitch_X" eval="true" value="0.00875"/>
<Parameter name="BigPixels_Pitch_Y" eval="true" value="0.0225"/>
</SpecPar>
```

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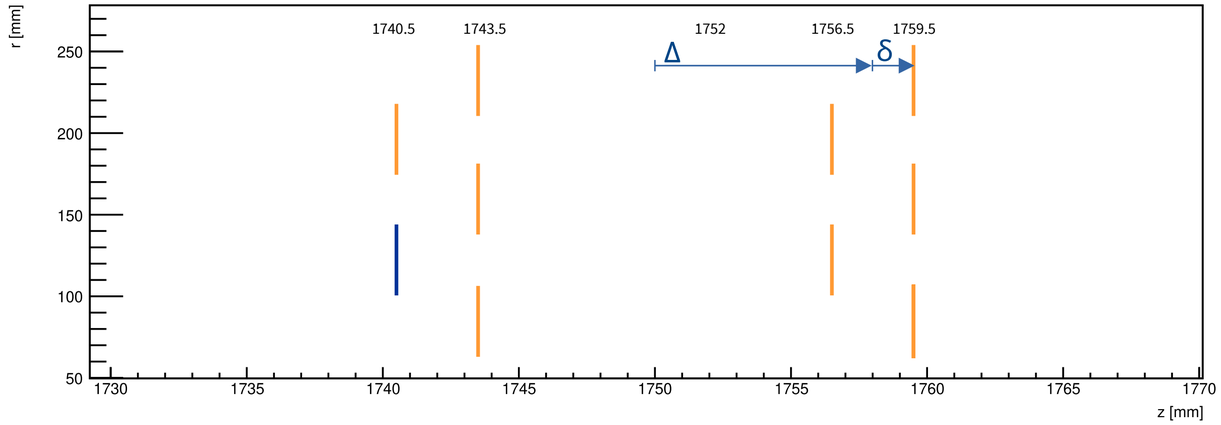
- **final value of TBPX gap**
  1.83 mm — realistic scenario

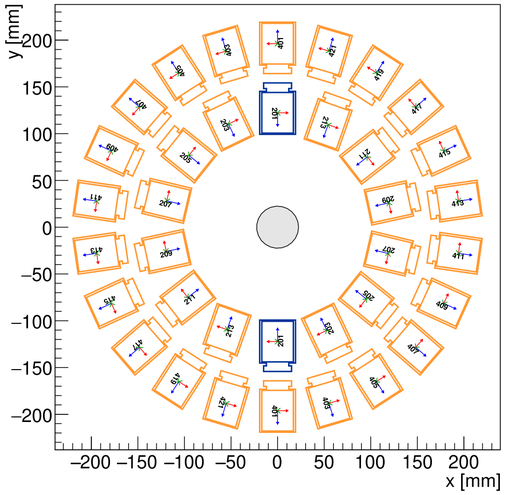


Final adjustments

Next baseline geometry in preparation: T40

- **big pixels in IT + OT** (PS-p sensors) **automated in tkLayout**
  IT: fully validated; **OT:** validation ongoing
- **final value of TBPX gap**
  1.83 mm — realistic scenario
- **synchronisation with the latest CAD designs [latest report by S.Mersi]**
  consistent sensor orientations, fixed placements of individual modules, overlap fixes, …





Final adjustments

Next baseline geometry in preparation: **T40**

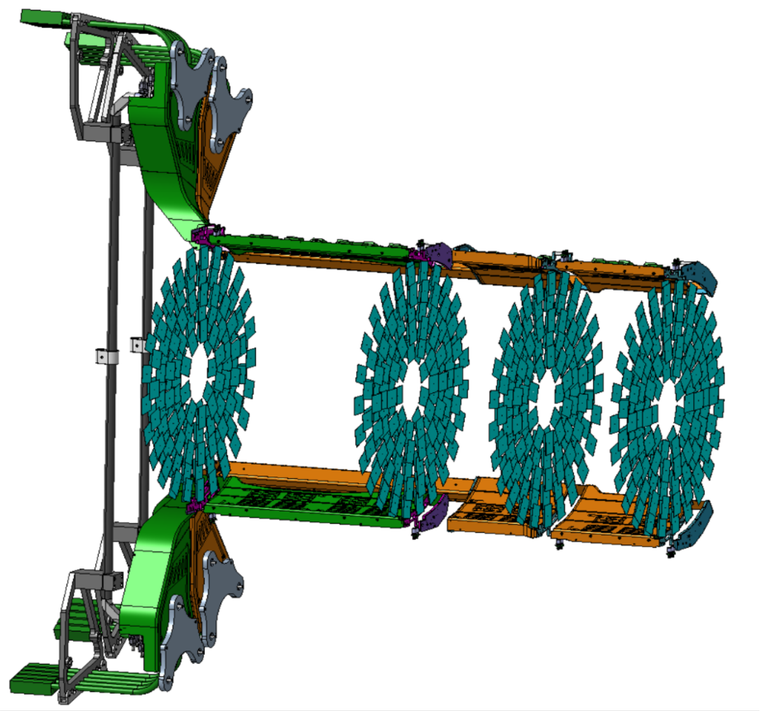
- **big pixels in IT + OT** (PS-p sensors) **automated in tkLayout**
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- **final value of TBPX gap**
  1.83 mm — realistic scenario
- **synchronisation with the latest CAD designs [latest report by S.Mersi]**
  consistent sensor orientations, fixed placements of individual modules, overlap fixes, …
  + **4 TFPX modules with adjusted R positions** (due to cabling)
  + **updated material definitions for TBPX** (discovered by A. Taliercio)

	$\|\eta\| \leq 0.8$	$0.8 < \|\eta\| \leq 1.6$	$1.6 < \|\eta\| \leq 2.4$	$2.4 < \|\eta\| \leq 3.2$	$3.2 < \|\eta\| \leq 4.0$
Current X_0	0.109	0.199	0.428	0.667	0.359
Updated X_0	0.112	0.203	0.434	0.676	0.365
Change	**2.8%**	**2.3%**	**1.4%**	**1.3%**	**1.6%**

Three levels of validation currently used:

- CAD ↔ tkLayout – comparing sensor positions

-	CAD	config file OT807_IT745 new	Difference	Units
Sensor surf. disc 1 to ref	1755.180			
Sensor plane disc 1.A.1	1755.255	1755.500	-0.245	mm
Sensor plane disc 1.A.2	1758.745	1758.500	0.245	mm
Middle plane disc 1A	1757.000	1757.000	0.000	mm
Sensor plane disc 1.B.1	1771.255	1771.500	-0.245	mm
Sensor plane disc 1.B.2	1774.745	1774.500	0.245	mm
Middle plane disc 1B	1773.000	1773.000	0.000	mm
Pitch disc 1A-2A	**240.300**			**mm**
Sensor plane disc 2.A.1	1995.555	1995.500	0.055	mm
Sensor plane disc 2.A.2	1999.045	1998.500	0.545	mm
Middle plane disc 2A	1997.300	1997.000	0.300	mm
Sensor plane disc 2.B.1	2011.555	2011.500	0.055	mm
Sensor plane disc 2.B.2	2015.045	2014.500	0.545	mm
Middle plane disc 2B	2013.300	2013.000	0.300	mm
Pitch disc 2A-3A	**259.700**			**mm**



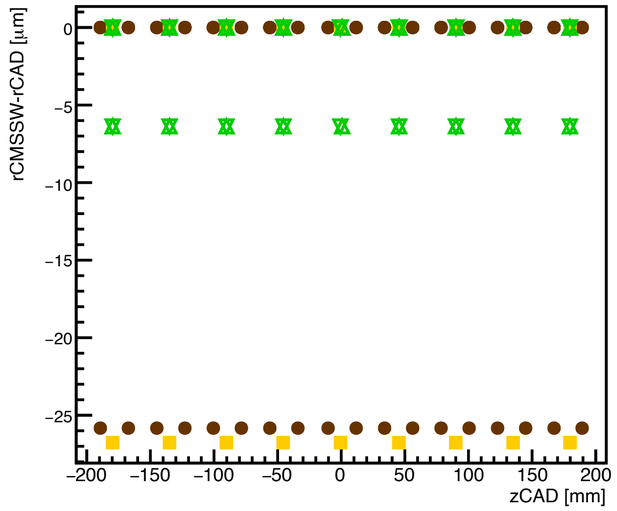
Recent comparison
by R.Tramontano, V.Guglielmi, Y.Herpin

Validations

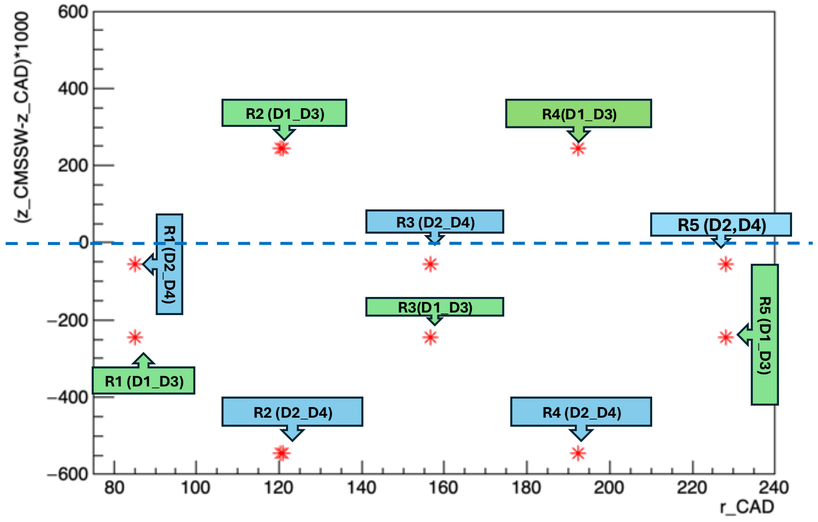
Three levels of validation currently used:

- CAD ↔ tkLayout — comparing sensor positions
- **CAD ↔ CMSSW — comparing sensor positions** (currently in IT only)

TBPX



TEPX



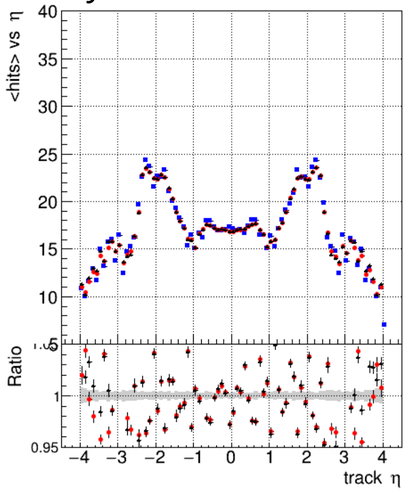
Latest comparison of the sensor positions (by F.Bashir, E.Migliore)

Validations

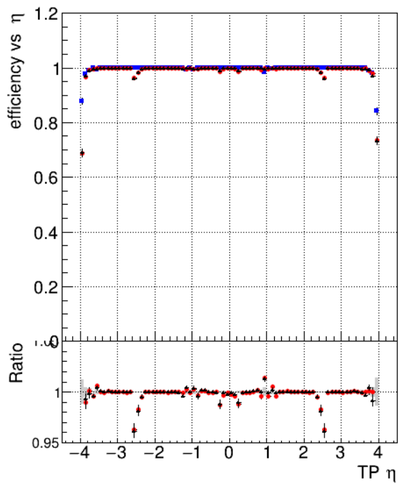
Three levels of validation currently used:

- CAD ↔ tkLayout – **comparing sensor positions**
- CAD ↔ CMSSW – **comparing sensor positions** (currently in IT only)
- **CMSSW** – **locally running standard Relase Validation workflows**
  MinBias + FourMuon + TTbar samples

by F.Fiore, G.Silva

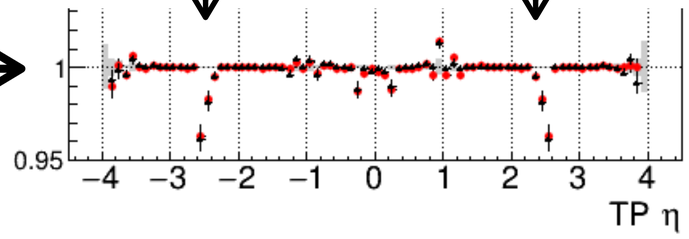


• • •



NO SMEAR
SMEAR

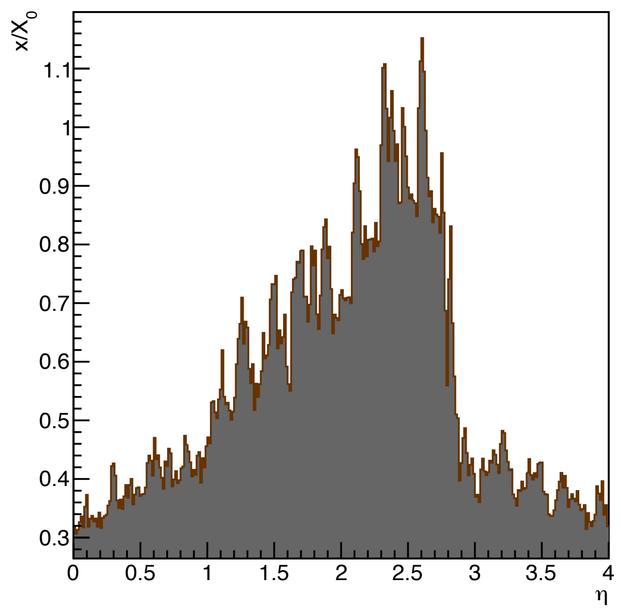
Bug in the tracking code



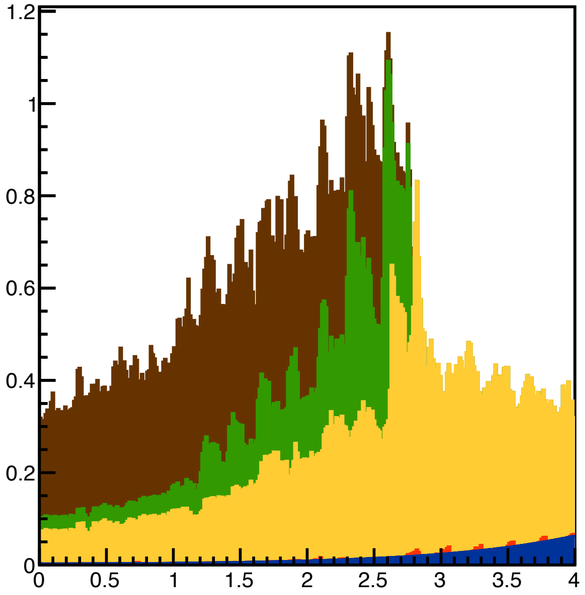
Material budget

All materials and dimensions defined in [tkLayout](#) (included in the exported XML files)

Radiation Length within Full Tracking Volume



Radiation Length by location in tracking volume



OT tracking Volume

OT Services + Support

IT tracking Volume

IT Services + Support

Beam pipe

Accurately reflects average material budget

High-granularity material budget

More precise description of passive materials needed in two places

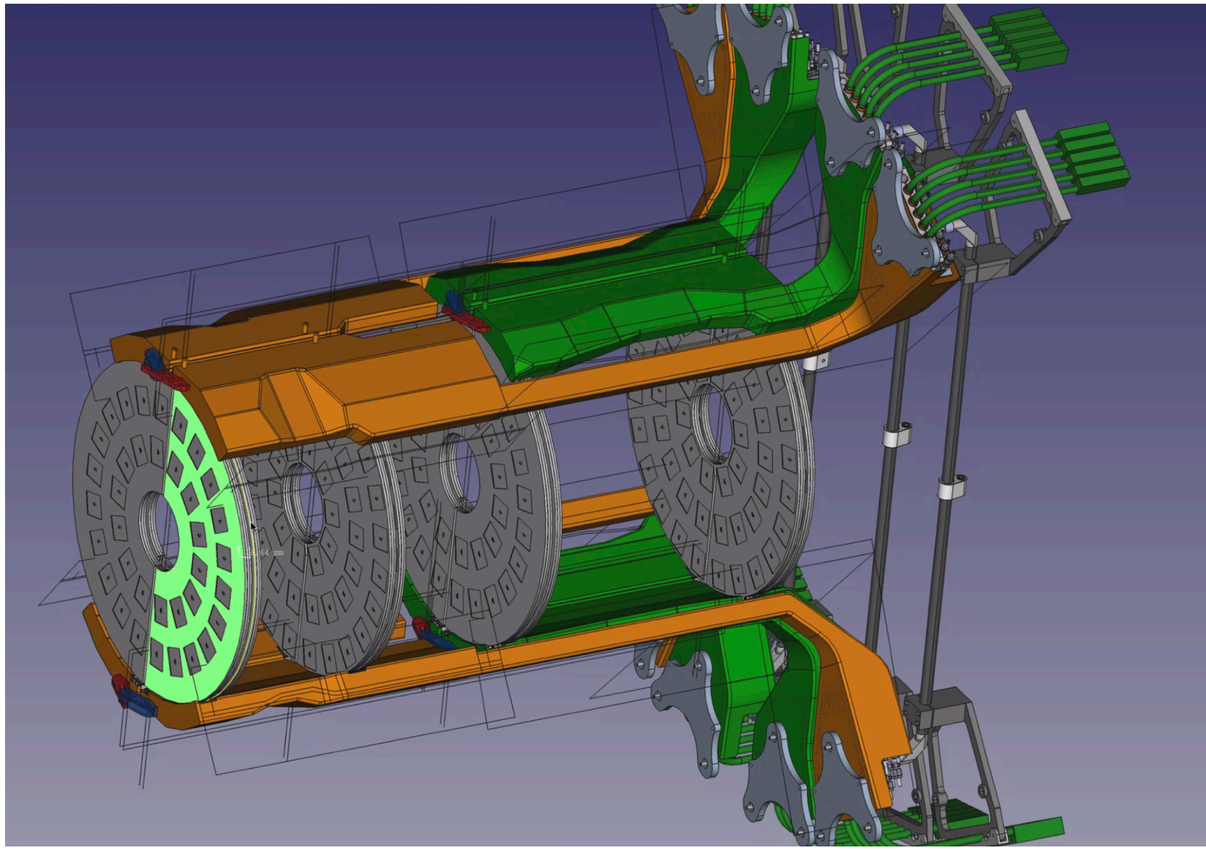
1. SIM geometry

Parametric shapes defined in tkLayout do not reflect local variations in material budget (cooling pipes, cables, etc.)

Will require a development of simplified CAD models of relevant passive materials

↳ to be imported straight into DD4hep

Long-term target



High-granularity material budget

More precise description of passive materials needed in two places

1. SIM geometry

Parametric shapes defined in tkLayout do not reflect local variations in material budget (cooling pipes, cables, etc.)

Will require a development of simplified CAD models of relevant passive materials

↳ to be imported straight into DD4hep

Long-term target

2. RECO geometry

Need for higher-granularity maps of material budget during track reconstruction

Crucial for Tracking POG

Fruitful discussion during the last [Geometry & Material budget meeting]

Tracking algorithms

Several distinct algorithms used at HLT (also used in offline tracking)



Material budget maps

Several distinct algorithms used at HLT (also used in offline tracking)

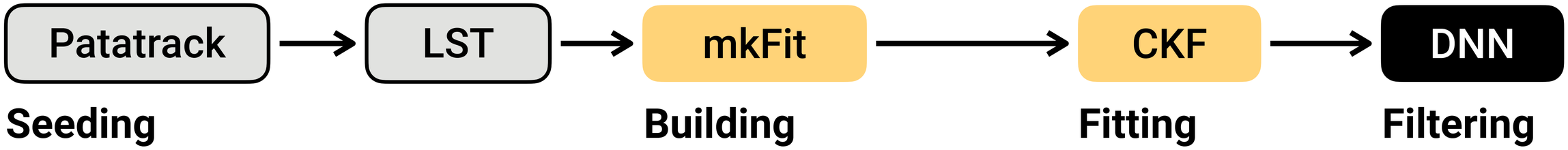


- one value for the whole tracker
  ↳ **fixed R-Z grid** (in progress)
- symmetric in ϕ
- used in the BrokenLine fit as a local density field

- material assigned to **sensitive volumes**
- granularity derived from Tracker Topology
  27 → **404 groups** (in progress)

Material budget maps

Several distinct algorithms used at HLT (also used in offline tracking)



Seeding

- one value for the whole tracker
  ↳ **fixed R-Z grid** (in progress)
- symmetric in ϕ
- used in the BrokenLine fit as a local density field

Building

- material assigned to **sensitive volumes**
- granularity derived from Tracker Topology
  27 → **404 groups** (in progress)

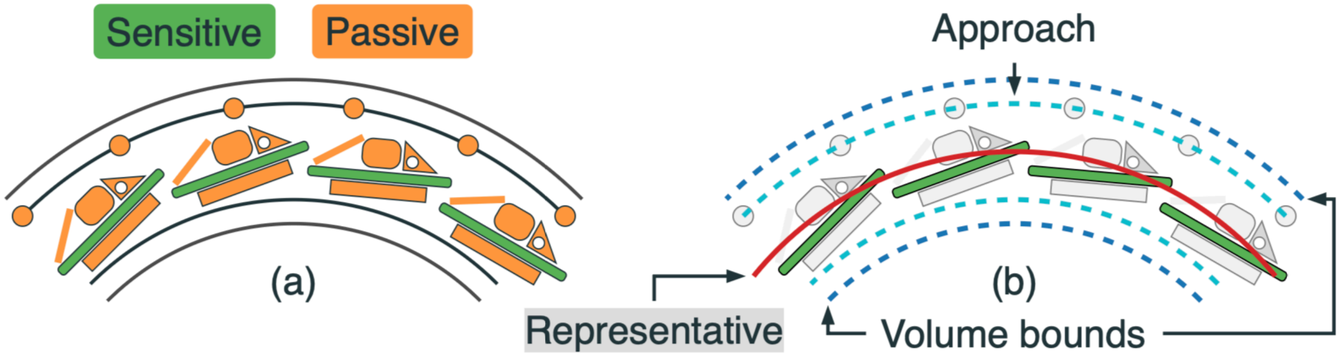
Fitting

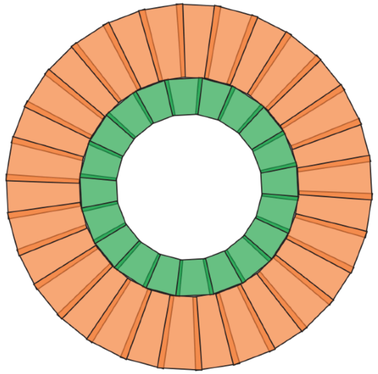
Filtering

Assignment to layer → layer transition
under study based on TGeo4mkFit

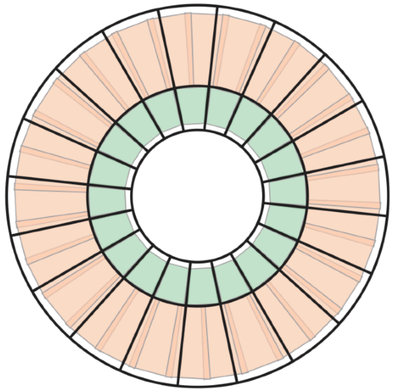
Sufficient for near-term developments of tracking algorithms

Alternative approach based on ACTS (A Common Tracking Software)





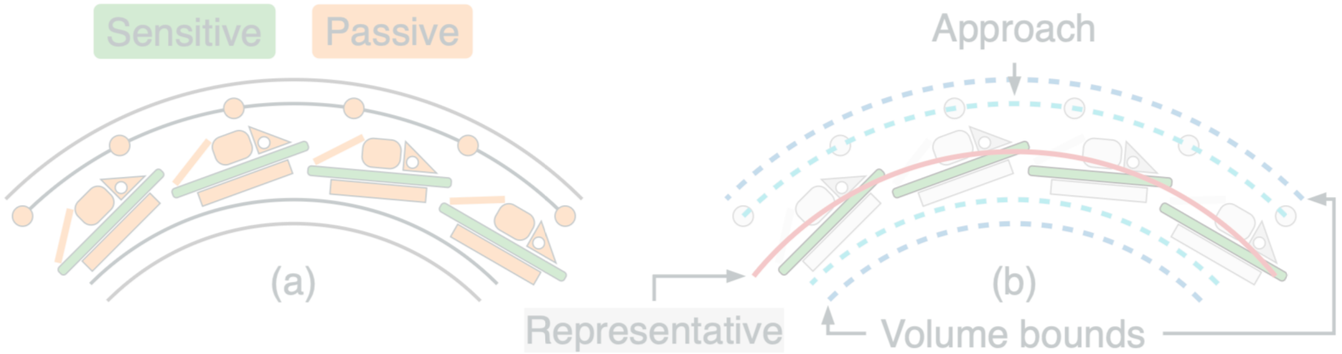
Sensor rings

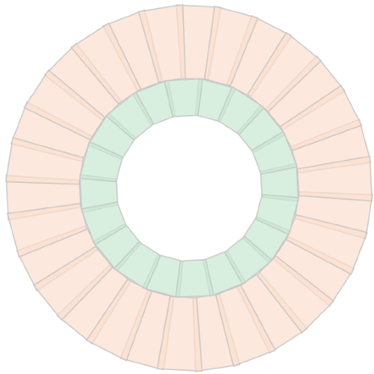


Map bins

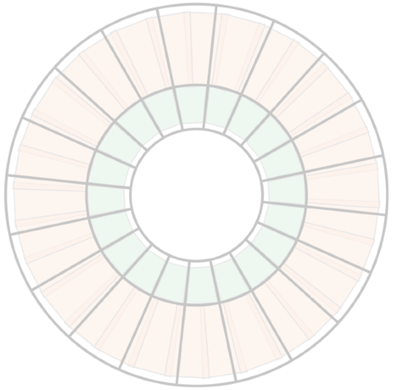
Material assigned to dedicated material layers
with a flexible binning structure

Alternative approach based on ACTS (A Common Tracking Software)





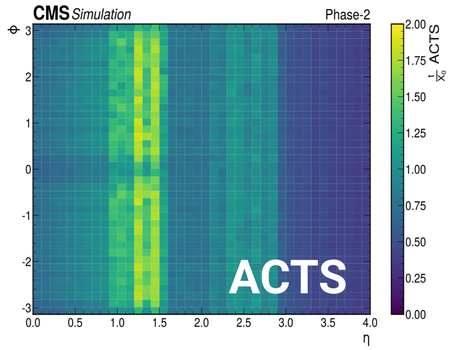
Sensor rings

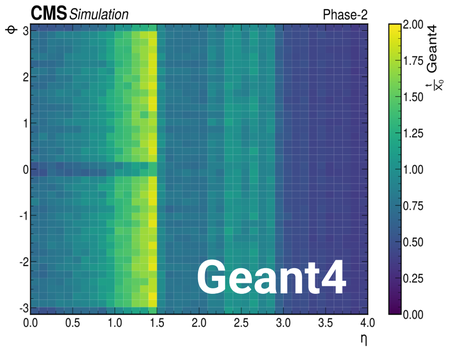


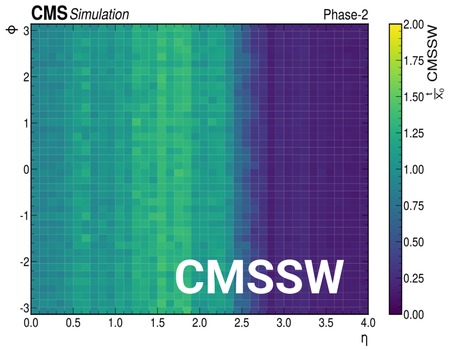
Map bins

Material assigned to dedicated material layers
with a flexible binning structure

Tested for track fitting in CMSSW
presented by L.Damenti



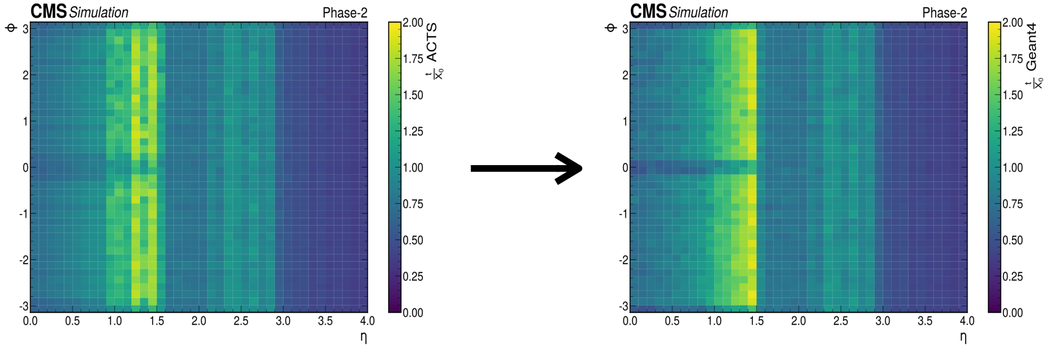


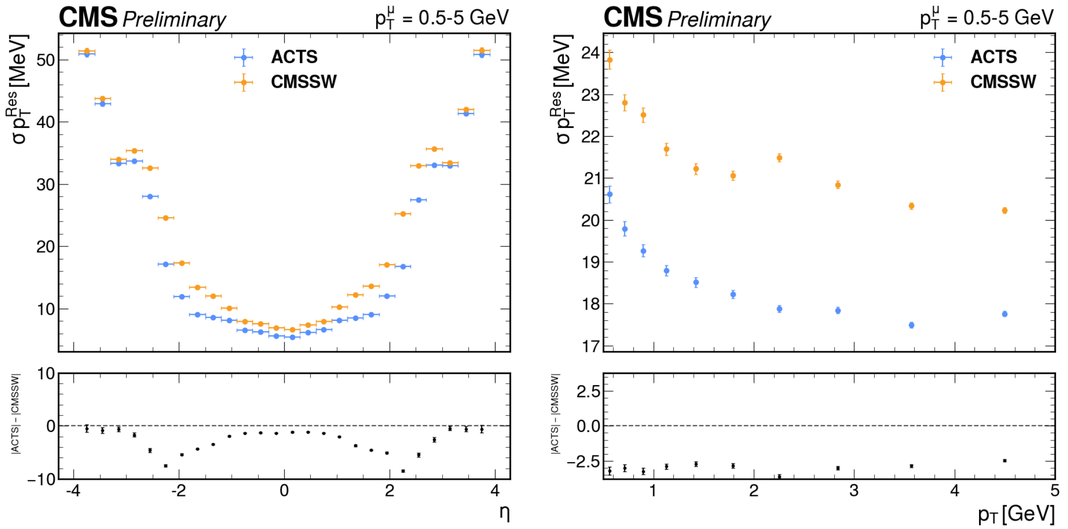


Material budget maps: ACTS method

First tests show clearly improved performance

Material description can be further optimised (better binning)





Proper integration into CMSSW requires significant effort

- **different propagator**
  taking into account CMS magnetic field
- **different energy-loss model**
- **interface with all tracking algorithms**

Long-term target

Current file structure of exported geometries is a bit cumbersome

- designed for progressive reuse of components from earlier geometries
- simpler structure is preferred for the converging geometry

dictRun4Geometry.py

```
'Geometry/TrackerCommonData/data/PhaseII/trackerParameters.xml',
'Geometry/TrackerCommonData/data/pixfwdCommon.xml',
'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_2021_02/pixfwd.xml',
'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_OT800_IT615_2022_10/pixbar.xml',
'Geometry/TrackerCommonData/data/trackermaterial.xml',
'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_2021_03/tracker.xml',
'Geometry/TrackerCommonData/data/PhaseII/OuterTracker616_2020_04/otst.xml',
'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_IT711_2023_05/pixel.xml',
'Geometry/TrackerCommonData/data/PhaseII/TiltedTracker404/trackerbar.xml',
'Geometry/TrackerCommonData/data/PhaseII/TiltedTracker404/trackerfwd.xml',
'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_2021_02/trackerStructureTopology.xml',
'Geometry/TrackerCommonData/data/PhaseII/Tracker_DD4hep_compatible_IT711_2024_02/pixelStructureTopology.xml',
'Geometry/TrackerSimData/data/PhaseII/Tracker_DD4hep_compatible_2021_02/trackersens.xml',
'Geometry/TrackerSimData/data/PhaseII/Tracker_DD4hep_compatible_IT711_2023_05/pixelsens.xml',
'Geometry/TrackerSimData/data/PhaseII/Tracker_DD4hep_compatible_2021_02/trackerProdCuts.xml',
'Geometry/TrackerSimData/data/PhaseII/Tracker_DD4hep_compatible_IT711_2023_05/pixelProdCuts.xml',
‘Geometry/TrackerSimData/data/trackerProdCutsBEAM.xml’,
'Geometry/TrackerRecoData/data/PhaseII/Tracker_DD4hep_compatible_IT711_2023_05/trackerRecoMaterial.xml',
```

Geometry organisation in CMSSW

More streamlined file structure for the next geometry: **T40**

- keep the split between **TrackerCommonData** - **TrackerSimData** - **TrackerRecoData**
- all documentation on the [tkLayout website]

```
'Geometry/TrackerCommonData/data/PhaseII/trackerParameters.xml',
'Geometry/TrackerCommonData/data/pixfwdCommon.xml',
'Geometry/TrackerCommonData/data/PhaseII/T40/pixfwd.xml',
'Geometry/TrackerCommonData/data/PhaseII/T40/pixbar.xml',
'Geometry/TrackerCommonData/data/trackermaterial.xml',
'Geometry/TrackerCommonData/data/PhaseII/T40/tracker.xml',
'Geometry/TrackerCommonData/data/PhaseII/T40/otst.xml',
'Geometry/TrackerCommonData/data/PhaseII/T40/pixel.xml',
‘Geometry/TrackerCommonData/data/PhaseII/TiltedTracker404/trackerbar.xml',
'Geometry/TrackerCommonData/data/PhaseII/TiltedTracker404/trackerfwd.xml',
'Geometry/TrackerCommonData/data/PhaseII/T40/trackerStructureTopology.xml',
'Geometry/TrackerCommonData/data/PhaseII/T40/pixelStructureTopology.xml',

'Geometry/TrackerSimData/data/PhaseII/T40/trackersens.xml',
'Geometry/TrackerSimData/data/PhaseII/T40/pixelsens.xml',
'Geometry/TrackerSimData/data/PhaseII/T40/trackerProdCuts.xml',
'Geometry/TrackerSimData/data/PhaseII/T40/pixelProdCuts.xml',
‘Geometry/TrackerSimData/data/trackerProdCutsBEAM.xml’,

‘Geometry/TrackerRecoData/data/PhaseII/T40/trackerRecoMaterial.xml',
```

Common configurations shared by all geometries

Summary

Tracker geometry is in a good shape
all major features implemented and validated

Minor adjustments being finalised → **T40**
sensor positioning, big pixels, material budget

Major future developents targeting material budget

1. SIM geometry

- simplified CAD → DD4hep

2. RECO geometry

- ACTS model → CMSSW

⚠ **Dedicated meeting**
reserved on Fridays →

16:00 → 17:20 **Geometry & material budget**

Convener: Nazar Bartosik (UPO e INFN Torino (IT))